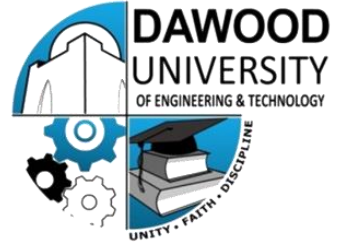


LECTURE – 03, 04 & 05

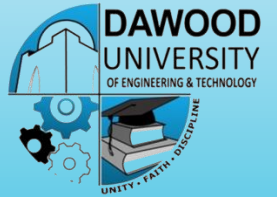
OOP (JAVA)



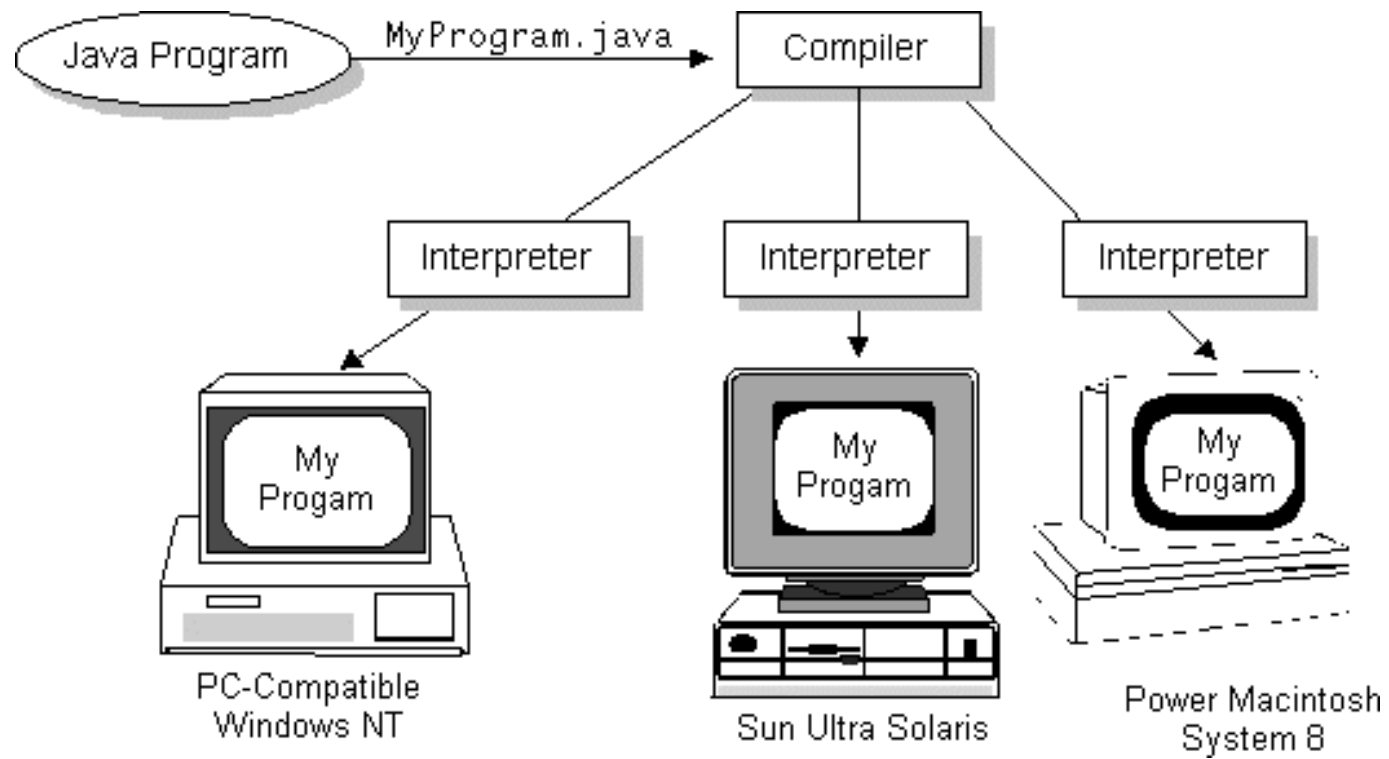
PREM KUMAR
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AGENDA



- Write Once Run Everywhere
- Compile Time
- JDK
- JRE
JVM

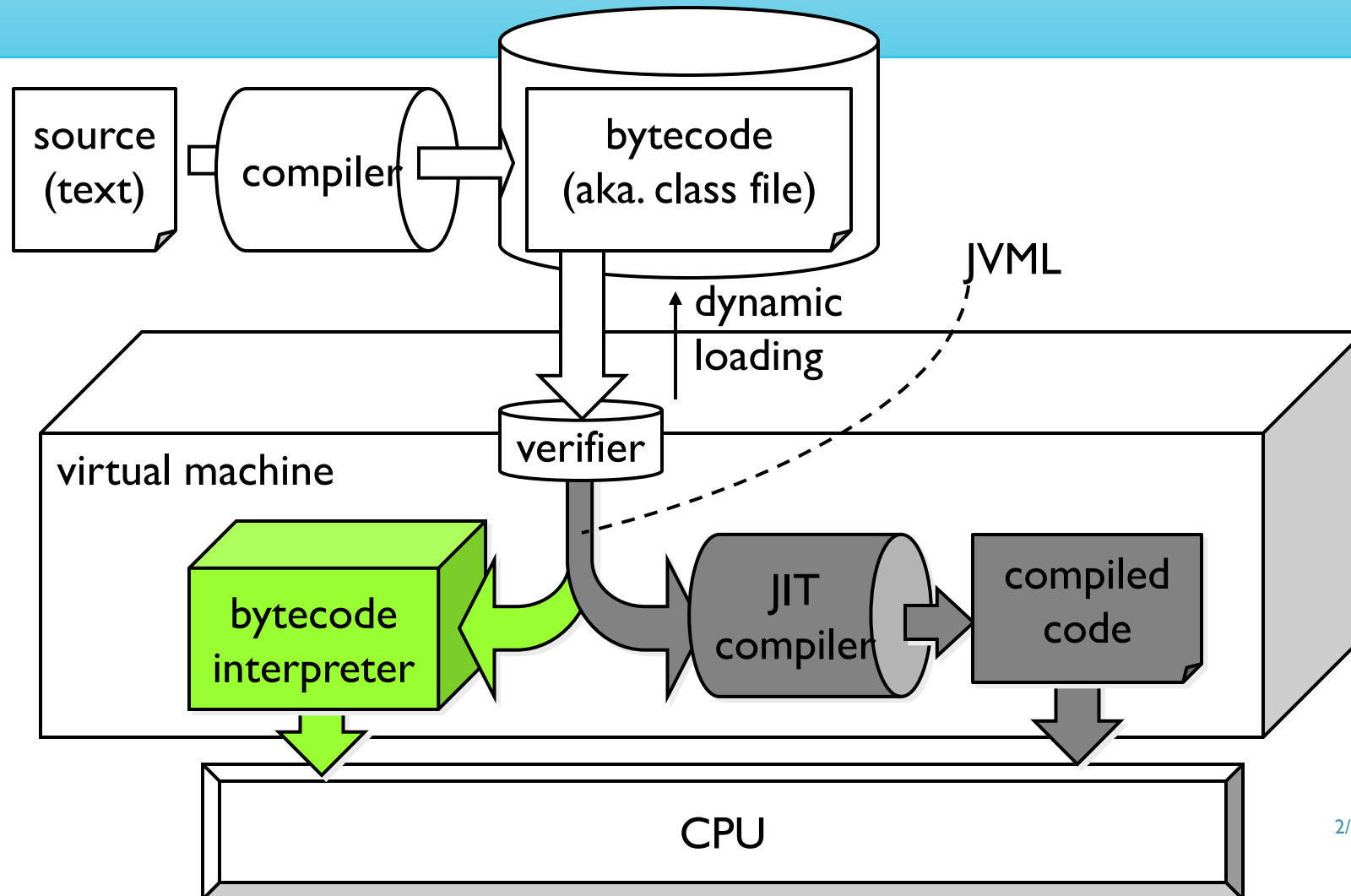


WRITE ONCE, RUN ANYWHERE

BYTECODE: FOOD FOR THE VM

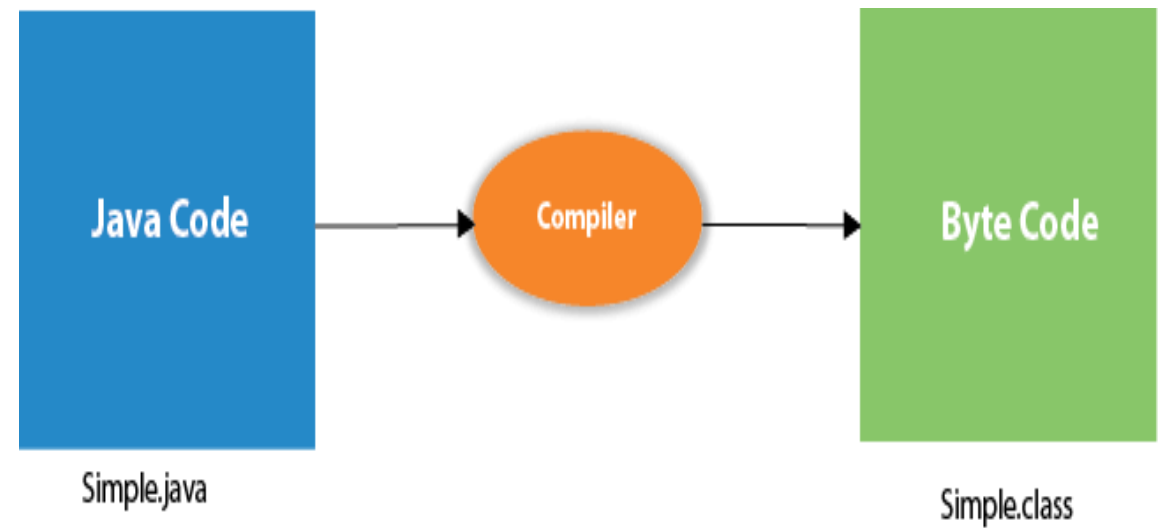
- For most languages, compilation produces machine code
- Java compilation produces “byte code”
 - Intermediate code readable by the VM
 - Transferable across the Internet as *applets*
- VM interprets BC into instructions
 - Partly responsible for performance lag
- Byte Code produced on any platform may be executed on any other platform which supports a VM

EXECUTION MODEL OF JAVA



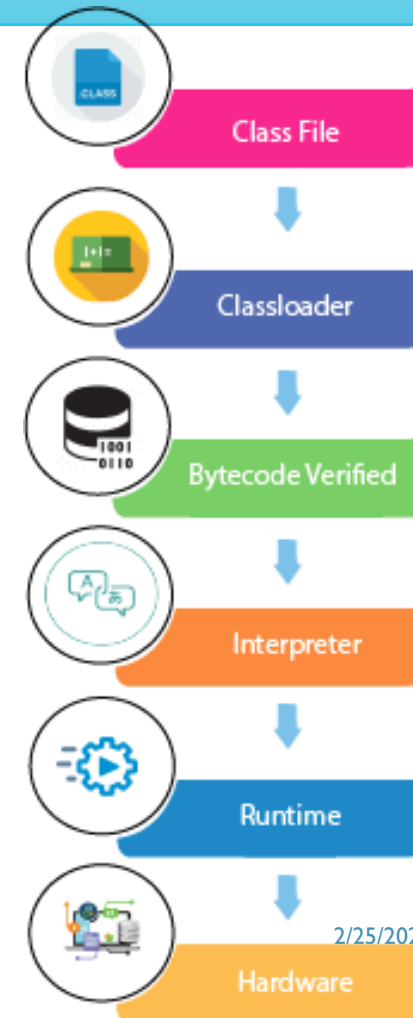
WHAT HAPPENS AT COMPILE TIME?

- At compile time, the Java file is compiled by Java Compiler (It does not interact with OS) and converts the Java code into bytecode.



WHAT HAPPENS AT RUNTIME?

- At runtime, the following steps are performed:
- **Classloader:** It is the subsystem of JVM that is used to load class files.
- **Bytecode Verifier:** Checks the code fragments for illegal code that can violate access rights to objects.
- **Interpreter:** Read bytecode stream then execute the instructions.



THE JIT



- *Just-In-Time* compiler
- Translates bytecode into machine code at runtime
 - Performance increase 10-30 times
- Now the default for most JVM's
 - Can be turned off if desired
 - JIT can apply statistical optimizations based on runtime usage profile

HOW TO SET PATH IN JAVA

- The path is required to be set for using tools such as javac, java, etc.
- If you are saving the Java source file inside the JDK/bin directory, the path is not required to be set because all the tools will be available in the current directory.
- However, if you have your Java file outside the JDK/bin folder, it is necessary to set the path of JDK.
- There are two ways to set the path in Java:
 1. Temporary
 2. Permanent

1) How to set the Temporary Path of JDK in Windows

To set the temporary path of JDK, you need to follow the following steps:

- Open the command prompt
- Copy the path of the JDK/bin directory
- Write in command prompt: set path=copied_path

For Example:

```
set path=C:\Program Files\Java\jdk1.6.0_23\bin
```

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December 2011

C:\Windows\system32\cmd.exe

Microsoft Windows [Version 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Sonoo>cd\

C:\>cd new

C:\new>javac Simple.java
'javac' is not recognized as an internal or external command,
operable program or batch file.

C:\new>set path=C:\Program Files\Java\jdk1.6.0_03\bin

C:\new>javac Simple.java

C:\new>java Simple
Hello Java

C:\new>

2) HOW TO SET PERMANENT PATH OF JDK IN WINDOWS

- For setting the permanent path of JDK, you need to follow these steps:
- Go to MyComputer properties -> advanced tab -> environment variables -> new tab of user variable -> write path in variable name -> write path of bin folder in variable value -> ok -> ok -> ok

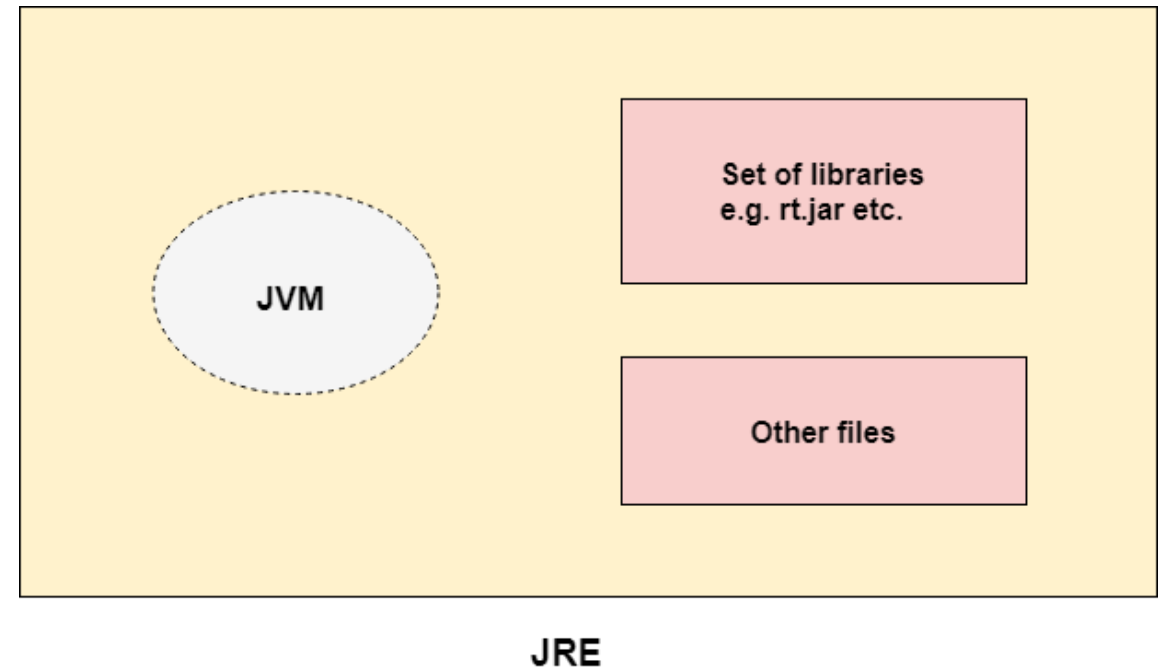
JVM



- JVM (Java Virtual Machine) is an abstract machine.
- It is called a virtual machine because it doesn't physically exist.
- It is a specification that provides a runtime environment in which Java bytecode can be executed.
- The JVM performs the following main tasks:
 - Loads code
 - Verifies code
 - Executes code
 - Provides runtime environment

JRE

- JRE is an acronym for Java Runtime Environment
- The Java Runtime Environment is a set of software tools which are used for developing Java applications.
- It is used to provide the runtime environment.
- It is the implementation of JVM.
- It physically exists.
- It contains a set of libraries + other files that JVM uses at runtime.
- The implementation of JVM is also actively released by other companies besides Sun Micro Systems.



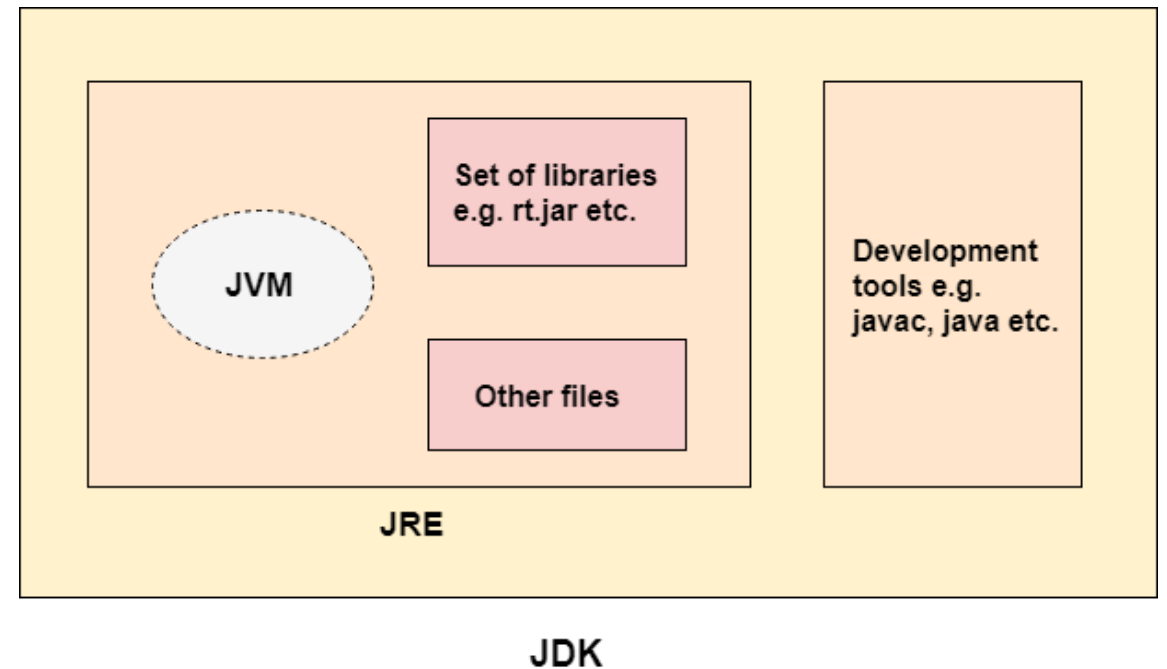
JDK



- JDK is an acronym for Java Development Kit.
- The Java Development Kit (JDK) is a software development environment which is used to develop Java applications and **applets**.
- It physically exists.
- It contains JRE + development tools.
- JDK is an implementation of any one of the below given Java Platforms released by Oracle Corporation:
 - Standard Edition Java Platform
 - Enterprise Edition Java Platform
 - Micro Edition Java Platform

JDK

- The JDK contains a private Java Virtual Machine (JVM) and a few other resources such as an interpreter/loader (java), a compiler (javac), an archiver (jar), a documentation generator (javadoc), etc. to complete the development of a Java Application.





Thanks 😊